

Personal Library & AI Research Engine — Full Capability Mindmap

Full-Stack Architectural Decomposition: Ingestion, BibTeX, Dual AI, Qdrant RAG, UI5, Security, and Governance

PERSONAL LIBRARY & AI RESEARCH ENGINE — COMPLETE FUNCTIONALITY MINDMAP

100% Comprehensive Capability Matrix: Spring Boot 4 • Spring AI 2 • UI5 Web Components • Qdrant Vector RAG • nginx Ingress • Keycloak OIDC

AGPL-3.0-or-later • Enterprise

5. VERSION CONTROL & LINEAGE

Audit Immutable

In-Place Version Overwriting

- Multipart replacement maintaining root GUID
- Increments versionNumber (V1 → V2 → Vn)

Version Lineage Chain

- Embedded versionHistory snapshot archiving
- Non-destructive historical preservation

Comprehensive Mutation Audit Trail

- Timestamps, editing user, changed fields
- Compliance logging in Keycloak/MongoDB

Rollback & Historical Asset Access

- Download previous version raw binary file
- Revert or fork version tree anytime

6. ENTERPRISE UI5 FRONTEND

SAP Horizon

ListReport & ObjectPage Floorplans

- Multi-field FilterBar, sorting, pagination
- Dynamic sticky header & KPI summary cards

Horizon & Horizon Dark Theming

- Real-time data-sap-theme toggling
- Shadow DOM UI5 component CSS injection

5 Localized Languages (i18n)

- English, German, French, Spanish, Romanian
- Instant reactive store translation updates

Observable Reactive State & Gateway

- Zero framework overhead (Core Component)
- Instant switch between REST and Mock driver

7. SECURITY & GOVERNANCE

Keycloak OIDC

Keycloak 24+ OIDC Authentication

- OAuth2 JWT Bearer token enforcement
- Single Sign-On (SSO) & session lifecycle

Role-Based Access Control (RBAC)

- LIBRARY_ADMIN (Full Write, Delete, Overwrite)
- CHIEF_RESEARCHER (AI benchmark & RAG)

Audit Event Logging & Defense

- Path traversal blocking & sanitized filenames
- Immutable access logs for audit compliance

GNU AGPL-3.0-or-later License

- Strongest network copyleft guarantee
- Transparent open research infrastructure

8. ARCHITECTURE & TOOLING

Full Stack

OpenAPI 3.0.3 & Swagger Spec

- Interactive REST endpoint documentation
- Downloadable JSON schema & test harness

Complete C4 & UML Diagrams

- System, Container, Component C4 layers
- Java Spring Boot + TS UI5 UML Class models

Multi-Format Visual Export Suite

- Vector SVG, 4K Ultra-HD PNG, Archival PDF
- PlantUML (.puml), Mermaid (.mmd), Draw.io

Self-Contained .ZIP & Docs

- Full repo export with scripts & models
- Javadoc + TypeDoc interactive docs

1. INGESTION & PIPELINE

Static BPMN

Multi-Format Document Ingestion

- PDF, DOCX, Markdown, LaTeX, Plain Text
- MIME detection & binary streaming

Drag-and-Drop File Upload UI

- Client-side BibTeX extraction preview
- Progress bars, status & error validation

Static BPMN 2.0 Export

- document-ingestion-rag.bpmn, no embedded e
- Served for external Camunda Modeler compat

Version-Isolated Physical Storage

- One directory per document version (no overwrite)
- Local/Cloud versioned physical disk archive

2. BIBTEX & LATEX ENGINE

AST Parser

AST Lexer & Regex Tokenizer

- Full parsing of @article, @book, etc.
- LaTeX accent sanitization (f{a}, f{e}, etc)

14 Structured Metadata Fields

- Title, Author, Year, Month, Booktitle, Journal
- DOI, URL, Volume, Pages, Publisher, Abstract

Interactive In-Browser BibTeX Editor

- Live syntax highlighting & schema check
- One-click clipboard citation copy

Multi-Criteria Faceted Filtering

- Filter by Year, Entry Type, Author
- MongoDB dynamic criteria execution

3. DUAL AI BENCHMARKING

Spring AI

Ollama Llama 3.3 (70B Instruct)

- Analytical synthesis & research critique
- Deep methodology decomposition

Ollama Mistral Large (2411)

- Executive summary & key bullet takeaways
- Rapid operational digest in sub-3.5 sec

Synchronized Comparator UI

- Dual-card side-by-side comparison
- Markdown rendered with math equations

Execution Telemetry & Caching

- Latency tracking (seconds/tokens)
- Immutable summary cache per version

4. VECTOR RAG & CHAT

Qdrant Vector

Semantic Text Chunking

- ~400-char paragraph-aligned chunks
- Header preservation & document GUID tag

Qdrant Vector Database

- library_embeddings collection
- Dense Cosine metric & HNSW sub-10ms index

Conversational Document Chat

- Multi-turn conversational memory
- Natural language prompt injection guard

Grounded Citation Drawer

- Verifiable excerpt text with relevance %
- Page markers & anti-hallucination badges

Functional Capability Matrix & Technical Specifications

Detailed Pillar-by-Pillar Breakdown of the 8 Core Architectural Subsystems

1. Ingestion & Document Pipeline

Multi-format document processing with automated orchestration.

- Binary formats: PDF, DOCX, Markdown, LaTeX, Plain Text with MIME validation.
- Static BPMN 2.0 export served for external Camunda Modeler compatibility; human review loop runs in-app.
- Version-isolated physical storage path per upload, preventing any prior version's bytes from being overwritten.
- Physical versioned file system with secure streaming endpoint delivery.

2. BibTeX & LaTeX Metadata Engine

AST-based academic citation parsing and manipulation.

- Full AST tokenization of @article, @book, @inproceedings, @techreport, @misc.
- 14 structured metadata attributes: title, author, year, month, DOI, URL, abstract, etc.
- LaTeX accent sanitization (converting \'{a}, \'{e} into UTF-8 standard characters).
- Live interactive browser BibTeX editor with schema validation and clipboard export.

3. Dual AI Models & Benchmarking

Spring AI multi-model comparative synthesis engine.

- Llama 3.3 (70B Instruct): Deep academic synthesis, critique, and methodology breakdown.
- Mistral Large (2411): High-speed executive summary and core operational takeaways.
- Fully local inference via the Ollama daemon — no cloud API keys required.
- Synchronized side-by-side comparator UI with latency and duration telemetry.

4. Vector RAG & Conversational Chat

Sub-10ms neural search and grounded conversational Q&A.

- Semantic chunking: paragraph-aligned chunks of roughly 400 characters, no fixed overlap window.
- Qdrant vector collection library_embeddings with Cosine distance metric & HNSW.
- Conversational assistant with multi-turn context memory and prompt guardrails.
- Interactive citation drawer with exact quote highlights and confidence scores.

5. Version Control & Lineage Tracking

Non-destructive in-place versioning and audit preservation.

- Multipart version overwriting that preserves the root document GUID.
- Version lineage via incrementing versionNumber and embedded versionHistory snapshots (no separate rootGuid field).
- Complete mutation audit history: timestamps, editing user, and modified fields.
- Historical document rollback and raw binary download for any prior version.

6. Enterprise UI5 Frontend Experience

SAP Fiori Horizon floorplans built on web components.

- Official @ui5/webcomponents v2 wrapping with zero external framework dependencies.
- Analytical ListReport with multi-field FilterBar and adaptive ObjectPage with KPI header.
- Dual theme support: SAP Horizon (Light) and SAP Horizon Dark.
- Internationalization (i18n): 5 Locales (English, German, French, Spanish, Romanian).

7. Security, Identity & Governance

Enterprise OAuth2/OIDC and role-based access control.

- Keycloak 24+ integration with JWT Bearer token authentication.
- Role-Based Access Control (RBAC): LIBRARY_ADMIN, CHIEF_RESEARCHER, VIEWER.
- Security audit logging for administrative mutations and file streaming operations.
- GNU AGPL-3.0-or-later licensing protecting free and open research collaboration.

8. Architecture, API & Tooling Ecosystem

Standards-compliant documentation and exporter tooling.

- OpenAPI 3.0.3 REST API specification and interactive Swagger schema viewer.
- Complete C4 Architecture diagrams and Java / TypeScript UML class diagrams.
- Automated multi-format exporter: SVG, Ultra-HD PNG, Archival PDF, Draw.io, Mermaid.
- Self-contained repository ZIP exporter with full source code, scripts, and models.